

The Dark-Fraction Ledger Is Geometry: One Universal Concentration Reproduces the Host-Mass Dependence, and the Lyman- α Forest Closes the Late-and-Puffy Component

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AI-assisted research programme; not peer reviewed

11 September 2026

Abstract

A MOND kernel on the de Sitter-locked acceleration scale $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ passes galaxies, clusters and the CMB together only if a real cold component is present with a fraction of its Λ CDM halo mass that rises with host mass: $\lesssim 0.105$ inside three disc scale lengths of a $1.2 \times 10^{10} M_\odot$ spiral, 0.14 inside 30 kpc of the Milky Way, 0.576 inside R_{500} of X-COP clusters, $\gtrsim 0.988$ at recombination (the repository’s necessity certificate; the trend is $f \propto M^{0.16}$). We show that the first three numbers are geometry: the three measurement radii sit at 0.50, 1.19 and 2.71 Λ CDM scale radii, so a component whose haloes all share one universal concentration c_* exhibits exactly such a rise. Fitting c_* to the three anchors gives $c_* = 0.40$ and $f = 0.086, 0.165, 0.704$, all within a factor 1.6 of the targets. The cost is that $c_* = 0.4$ lies far below the concentration $c \simeq 3\text{--}4$ of any halo that has just collapsed: the component would have to have no structure below roughly a megaparsec, i.e. a linear spectrum cut at $k_{\text{cut}} \lesssim 1 h \text{ Mpc}^{-1}$. Whether nonlinear top-down fragmentation could regenerate the Lyman- α forest power from such a spectrum was an open door of the programme; we close it with a phase-matched particle-mesh comparison: at $k = 5 h \text{ Mpc}^{-1}$ and $z = 2.2$ the regenerated power is 0.04, 0.45, 0.81 of Λ CDM’s for $k_{\text{cut}} = 1, 2, 4 h \text{ Mpc}^{-1}$, against a forest tolerance of 0.9, so the forest requires $k_{\text{cut}} > 4$ (structure intact down to $\sim 2 \times 10^{11} M_\odot$) while the puffy profile requires $k_{\text{cut}} \lesssim 1$. With the acoustic route closed by the clock stability theorem and the decay/kick route by a lifetime pincer, the three physical routes to a host-mass-dependent depletion are now each computed to failure, and the ledger’s geometric shape is the exact target that any future mechanism must reproduce.

1 The ledger

The repository [1] carries a Lean-certified necessity result: any completion of the de Sitter-locked MOND kernel that passes rotation curves, clusters, the CMB and the Lyman- α forest must contain a real, clustering cold component whose retained fraction f of the Λ CDM halo mass rises with host mass. Its computed anchors are listed in Table 1; the trend between spirals and clusters is $f \propto M^{0.16}$. The acoustic route to such a depletion (a sound speed) was closed by the clock stability theorem [2]: a fluid soft enough for the forest clusters inside galaxies. The kinetic route (two-body

decay or velocity kicks) is closed by a lifetime pincer, forest $\tau \geq 41$ Gyr against galaxies $\tau \leq 20$ Gyr. This note addresses what remains.

Table 1: The ledger’s anchors and their radii in Λ CDM scale-radius units ($h = 0.6736$; concentrations from [4]).

host	$M_{200} [M_\odot]$	$R_{200} [\text{Mpc}]$	$c_{\Lambda\text{CDM}}$	radius	r/r_s	target f
SPARC spiral, $1.2 \times 10^{10} M_\odot$ baryons	3×10^{11}	0.142	9.4	$3R_d = 7.5 \text{ kpc}$	0.50	≤ 0.105
Milky Way (Gaia DR3 decline)	10^{12}	0.212	8.4	30 kpc	1.19	0.14
X-COP cluster	10^{15}	2.116	4.2	$R_{500} = 1.38 \text{ Mpc}$	2.71	0.576

2 Geometry: one universal concentration

Let the real component’s haloes be NFW [3] with a universal concentration c_* and the same M_{200} as the Λ CDM halo. With $m(x) = \ln(1+x) - x/(1+x)$, the retained fraction inside radius r is

$$f(r) = \frac{m(rc_*/R_{200})/m(c_*)}{m(rc_{\Lambda\text{CDM}}/R_{200})/m(c_{\Lambda\text{CDM}})}. \quad (1)$$

Because the measurement radii of Table 1 climb from $0.5 r_s$ to $2.7 r_s$, f rises with host mass for any $c_* < c_{\Lambda\text{CDM}}$: a low-concentration profile is depleted where one looks deep inside the scale radius and nearly complete where one looks outside it. Table 2 gives f for a range of c_* ; minimising $\sum(\ln f - \ln f_{\text{target}})^2$ over $c_* \in [0.3, 6]$ gives $c_* = 0.40$.

Table 2: Retained fraction at the ledger’s radii for a universal concentration c_* .

host	target	$c_* = 0.4$	1	1.5	2	3	4
SPARC spiral ($0.50 r_s$)	0.105	0.086	0.135	0.180	0.227	0.325	0.428
Milky Way ($1.19 r_s$)	0.14	0.165	0.244	0.310	0.373	0.494	0.606
X-COP cluster ($2.71 r_s$)	0.576	0.704	0.798	0.852	0.894	0.953	0.994

At $c_* = 0.40$ all three anchors are reproduced within a factor 1.6 (ratios 0.81, 1.18, 1.22). The ledger’s mass dependence therefore needs no mass-dependent physics: it is what one universal, extremely extended profile looks like through three different windows.

3 What it costs

A concentration of 0.4 is not a property any collapsed halo can have. In the standard picture $c \simeq 4(1+z_c)$ at $z = 0$ [5, 6], so a halo that has only just collapsed has $c \simeq 3$ –4; haloes with $c < 3$ are unrelaxed systems still in formation. A component with $c_* = 0.4$ has a scale radius of $2.5 R_{200}$: it possesses essentially no structure below roughly a megaparsec, which requires its linear power to be cut at $k_{\text{cut}} \lesssim 1 h \text{ Mpc}^{-1}$. That is precisely the “nonlinear top-down fragmentation” loophole the programme had left open: could gravitational collapse of the surviving large-scale modes regenerate the small-scale power the forest measures by $z \simeq 2$ –3? Regeneration of small-scale power from truncated spectra is a known phenomenon at late times [7]; the question here is quantitative and at high redshift.

4 The top-down test

We evolve a cold component with a comoving particle-mesh code (the scheme of the repository’s L176 baseline, validated there against halofit): box $25 h^{-1}\text{Mpc}$, mesh 192^3 , 128^3 particles, Zel’dovich initial conditions at $z = 49$ from the CLASS linear spectrum, leapfrog in $\ln a$ with step 0.02. Four runs share the same random phases: ΛCDM and three spectra multiplied by $\exp[-(k/k_{\text{cut}})^2]$ with $k_{\text{cut}} = 1, 2, 4 h\text{Mpc}^{-1}$. The absolute ΛCDM power in this small box is 0.5–0.85 of halofit for $1 \leq k \leq 5 h\text{Mpc}^{-1}$ at $z = 3$ and 2.2 (the box lacks modes below $0.25 h\text{Mpc}^{-1}$ and the mesh limits the force resolution); this is a computed limitation, so only the phase-matched ratios between runs are read.

Table 3: Regenerated power $P(k; k_{\text{cut}})/P(k; \Lambda\text{CDM})$, same phases, $\pm 15\%$ bands.

$k_{\text{cut}} [h/\text{Mpc}]$	z	$k = 0.5$	1	2	3	5	8	linear input at $k = 5$
1	3.0	0.77	0.44	0.12	0.043	0.012	0.004	10^{-11}
1	2.2	0.79	0.48	0.18	0.091	0.043	0.025	10^{-11}
2	3.0	0.94	0.83	0.59	0.44	0.30	0.23	0.002
2	2.2	0.95	0.85	0.67	0.56	0.45	0.41	0.002
4	3.0	0.99	0.96	0.88	0.82	0.73	0.67	0.21
4	2.2	0.99	0.97	0.91	0.87	0.81	0.77	0.21

Regeneration is real: for $k_{\text{cut}} = 2$ the power at $k = 5 h\text{Mpc}^{-1}$ climbs from a linear input of 0.002 to 0.45 of ΛCDM ’s by $z = 2.2$. It is far from sufficient. With the forest tolerance used throughout the programme (dark-matter power at $k = 5 h\text{Mpc}^{-1}$, $z = 2$ –3, within 10%), the forest requires $k_{\text{cut}} > 4 h\text{Mpc}^{-1}$, i.e. the component’s structure intact down to haloes of $\sim 2 \times 10^{11} M_{\odot}$ (the mass inside a sphere of radius π/k_{cut} is 1.4×10^{13} , 1.7×10^{12} , $2.2 \times 10^{11} M_{\odot}$ for $k_{\text{cut}} = 1, 2, 4$). The puffy profile of Section 3 requires $k_{\text{cut}} \lesssim 1$. The late-and-puffy component is closed by the forest.

5 Consequences

Three physical routes could give a real component a retained fraction that rises with host mass: pressure support, kinetic removal, and late collapse. Each is now computed to failure on the programme’s own gates – pressure by the clock stability theorem and the Jeans argument [2], kinetic removal by the forest–galaxy lifetime pincer, late collapse by Table 3. What survives is the shape: the ledger is what a universal $c_* \simeq 0.4$ profile looks like at 0.5, 1.2 and 2.7 scale radii. Any future mechanism must therefore deplete the component inside a few scale radii of every halo while leaving its power on megaparsec scales untouched at $z \geq 2$; the geometric reading makes that requirement exact and testable.

6 What this does not claim

No statement is made that the ledger’s anchors are themselves beyond revision (they are the repository’s computed numbers with stated assumptions: halo masses from a stellar-to-halo relation, Dutton–Macciò concentrations, the radii of Table 1), nor that NFW is the right shape for an unrelaxed component. The particle-mesh comparison has a 50% absolute floor and no hydrodynamics;

the forest criterion is the linear-power proxy used in the repository’s earlier forest tests, not a flux-power likelihood. No claim is made about mechanisms outside the three classes named.

7 Reproducibility

`fable_independent_2026/L187_late_puffy_component_topdown.py` with its committed output and `L187_results.json` (commit `76f84c4dc`); the necessity certificate and ledger in `fable_independent_2026/FINDINGS.md` (L166, L172) and `lean_2026/Mondlean.lean`; the particle-mesh scheme in `L176_framework_pk_baseline.py`.

References

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